

```
In [1]: import pandas as pd
        from sklearn.model_selection import train_test_split
        from sklearn.preprocessing import StandardScaler
        from sklearn.neural_network import MLPClassifier
        from sklearn.metrics import accuracy_score
```

```
In [2]: dataset = pd.read_csv("letter-recognition.data", sep = ",")
```

```
In [3]: X = dataset.iloc[:, 1 : 17]
        Y = dataset.select_dtypes(include = [object])
```

Out:

	T
0	I
1	D
2	N
3	G
4	S
...	...
19994	D
19995	C
19996	T
19997	S
19998	A

19999 rows × 1 columns

```
In [4]: X_train, X_validation, Y_train, Y_validation = train_test_split(X, Y, test_size = 0.2
        0, random_state = 10)
```

```
In [5]: scaler = StandardScaler()  
        scaler.fit(X_train)
```

Out:

```
StandardScaler()
```

```
In [6]: X_train = scaler.transform(X_train)  
        X_validation = scaler.transform(X_validation)
```

```
In [7]: mlp = MLPClassifier(hidden_layer_sizes = (250, 300), max_iter = 1000000, activation =  
        'logistic')
```

```
In [8]: from yellowbrick.classifier import confusion_matrix  
        cm = confusion_matrix(mlp,X_train,Y_train, X_validation, Y_validation, classes="A,B,  
        C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T,U,V,W,X,Y,Z".split(','))
```

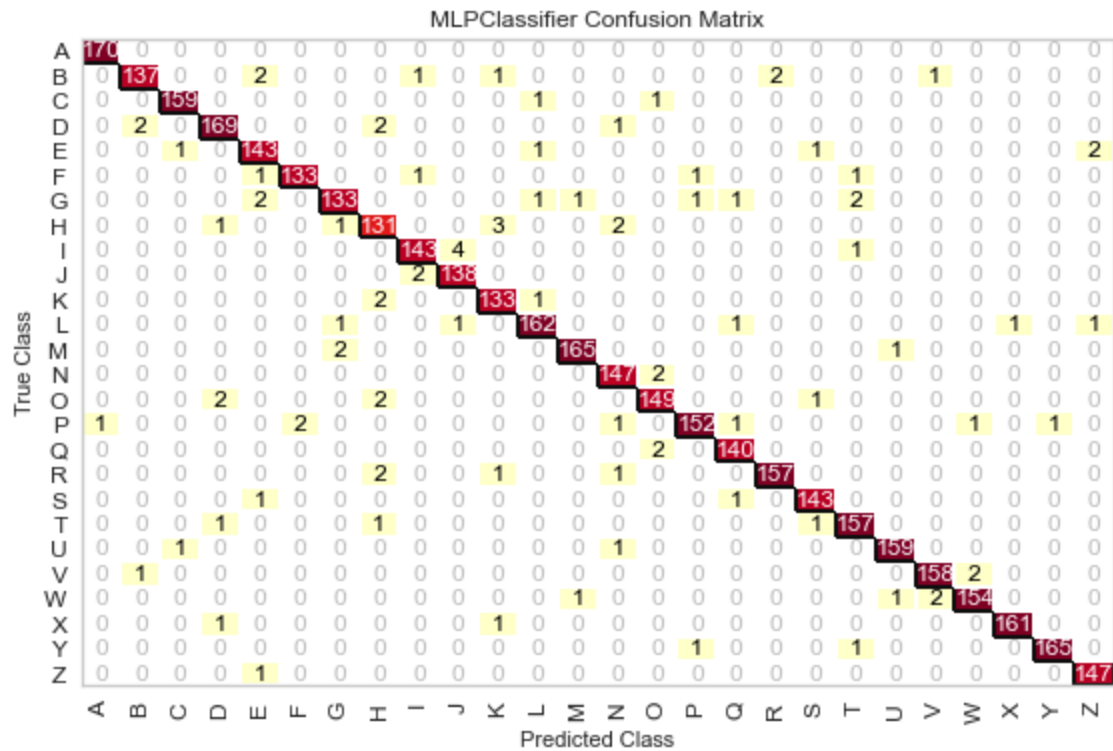
Out:

```
c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages\sklearn\netwo  
rk\_multilayer_perceptron.py:1109: DataConversionWarning: A column-vector y was passed when a  
1d array was expected. Please change the shape of y to (n_samples, ), for example using ravel  
().
```

```
    y = column_or_1d(y, warn=True)
```

```
c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages\sklearn\preprocessin  
g\_label.py:115: DataConversionWarning: A column-vector y was passed when a 1d array was expe  
cted. Please change the shape of y to (n_samples, ), for example using ravel().
```

```
    y = column_or_1d(y, warn=True)
```



In [9]: # !pip install yellowbrick

Out:

Collecting yellowbrick

WARNING: Ignoring invalid distribution -illow (c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages)
 WARNING: Ignoring invalid distribution -illow (c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages)

Downloading yellowbrick-1.5-py3-none-any.whl (282 kB)

----- 282.6/282.6 kB 1.2 MB/s eta 0:00:00

Requirement already satisfied: matplotlib!=3.0.0,>=2.0.2 in c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages (from yellowbrick) (3.5.1)

Requirement already satisfied: scipy>=1.0.0 in c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages (from yellowbrick) (1.7.3)

Requirement already satisfied: scikit-learn>=1.0.0 in c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages (from yellowbrick) (1.0.2)

Requirement already satisfied: numpy>=1.16.0 in c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages (from yellowbrick) (1.21.5)

Requirement already satisfied: cyclo>=0.10.0 in c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages (from yellowbrick) (0.11.0)

```

Requirement already satisfied: fonttools>=4.22.0 in c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages (from matplotlib!=3.0.0,>=2.0.2->yellowbrick) (4.29.1)
Requirement already satisfied: kiwisolver>=1.0.1 in c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages (from matplotlib!=3.0.0,>=2.0.2->yellowbrick) (1.3.2)
Requirement already satisfied: packaging>=20.0 in c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages (from matplotlib!=3.0.0,>=2.0.2->yellowbrick) (21.3)
Requirement already satisfied: pillow>=6.2.0 in c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages (from matplotlib!=3.0.0,>=2.0.2->yellowbrick) (9.5.0)
Requirement already satisfied: pyparsing>=2.2.1 in c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages (from matplotlib!=3.0.0,>=2.0.2->yellowbrick) (3.0.7)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages (from matplotlib!=3.0.0,>=2.0.2->yellowbrick) (2.8.2)
Requirement already satisfied: joblib>=0.11 in c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages (from scikit-learn>=1.0.0->yellowbrick) (1.1.0)
Requirement already satisfied: threadpoolctl>=2.0.0 in c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages (from scikit-learn>=1.0.0->yellowbrick) (3.1.0)
Requirement already satisfied: six>=1.5 in c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.0.0,>=2.0.2->yellowbrick) (1.16.0)
Installing collected packages: yellowbrick
Successfully installed yellowbrick-1.5

```

```
In [10]: cm.fit(X_train, Y_train.values.ravel())
```

Out:

```

ConfusionMatrix(ax=<AxesSubplot:title={'center': 'MLPClassifier Confusion Matrix'}, xlabel='Predicted Class', ylabel='True Class'>,
                classes=['A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J', 'K',
                          'L', 'M', 'N', 'O', 'P', 'Q', 'R', 'S', 'T', 'U', 'V',
                          'W', 'X', 'Y', 'Z'],
                cmap=<matplotlib.colors.ListedColormap object at 0x0000015FD8E45F88>,
                estimator=MLPClassifier(activation='logistic',
                                         hidden_layer_sizes=(250, 300),
                                         max_iter=1000000))

```

```
In [11]: cm.score(X_validation, Y_validation)
```

Out:

```

c:\users\lenovo\appdata\local\programs\python\python37\lib\site-packages\sklearn\preprocessing\_label.py:115: DataConversionWarning: A column-vector y was passed when a 1d array was expected

```

```
cted. Please change the shape of y to (n_samples, ), for example using ravel().  
y = column_or_1d(y, warn=True)
```

```
0.97625
```

```
In [12]: predictions = cm.predict(X_validation)  
predictions
```

Out:

```
array(['C', 'I', 'H', ..., 'H', 'B', 'W'], dtype='<U1')
```

```
In [13]: print("Accuracy: ", accuracy_score(Y_validation, predictions))
```

Out:

```
Accuracy:  0.97625
```